DECISION GUIDE · BAAVANA PODUGU

Machine Learning or Deep Learning?

A one-page guide for choosing the right approach before you start an AI project.

Traditional Machine Learning

People choose the features. Algorithms such as logistic regression, decision trees, and gradient-boosted trees learn from them.

Think: giving a student a checklist.

Deep Learning

A neural network with many layers learns the features directly from raw data such as images, audio, or text.

Think: teaching a student to read and letting them learn from experience.

Answer five questions

Question	If yes, lean toward…	Why
1. Is the data structured (rows and columns with clear meaning)?	Machine learning	Tree-based models perform very well on tabular data and train quickly.
2. Can experts name the factors that matter?	Machine learning	Known factors can be built into features directly.
3. Is the input raw and unstructured (images, audio, free text)?	Deep learning	Networks learn patterns from raw input that are hard to define by hand.
4. Do you have a very large labeled dataset and GPU computing?	Deep learning	Deep networks need lots of data and compute to reach their potential.
5. Must every prediction be explained to people?	Machine learning	Simpler models make it easier to show why a prediction was made.

Rule of thumb: start with the simplest model that solves the problem well. Move to deep learning when the data is unstructured, the patterns are too complex to define, and you have the data and computing to support it.

Two real-world examples

Customer churn prediction → Machine learning. A telecom company used tree-based models on structured customer records, including social network features, to predict which customers would leave. The best model (XGBoost) reached an AUC of 93.3% (Ahmad et al., 2019).

Diabetic eye disease screening → Deep learning. Google trained a convolutional neural network on about 128,000 graded retinal photos. It detected referable diabetic retinopathy with accuracy comparable to ophthalmologists (Gulshan et al., 2016).

Ahmad, A. K., Jafar, A., & Aljoumaa, K. (2019). Customer churn prediction in telecom using machine learning in big data platform. *Journal of Big Data, 6*(1), Article 28. https://doi.org/10.1186/s40537-019-0191-6

Gulshan, V., et al. (2016). Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs. *JAMA, 316*(22), 2402–2410. https://doi.org/10.1001/jama.2016.17216